3.SOFTWARE PEER REVIEWS & INSPECTIONS

Module Name: Review & Inspection

Test Case ID: str_id_03

Webpage URL: https://swehb.nasa.gov/spaces/SWEHBVB/pages/32604573/SWE-087

- **Scenario ID: 1a**

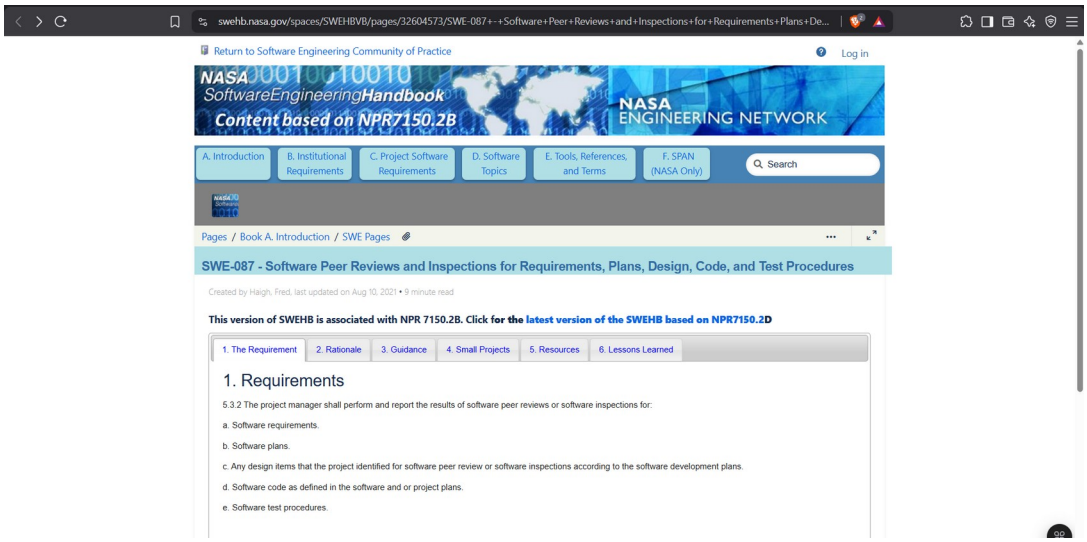
  Introduction & Purpose

This document presents a complete analysis of NASA SWE-087 (Software Peer Reviews and Inspections for Requirements, Plans, Design, Code, and Test Procedures) performed from the perspective of a Software Tester. All screenshots were captured from the official NASA Software Engineering Handbook page and are embedded below with explanations suitable for Tutorial-3 submission.

The goal is to understand formal requirement reviews, walkthroughs, peer reviews and inspections, their differences, roles, process steps, benefits and the critical value of performing them early (especially on requirements).

- **Scenario ID: 2b**

  Main Requirement Statement



The core mandatory requirement (NPR 7150.2B §5.3.2) states that the project manager shall perform and report the results of software peer reviews or software inspections for:

- a. Software requirements
- b. Software plans
- c. Any design items identified for peer review/inspection according to the software development plans
- d. Software code as defined in the software and/or project plans
- e. Software test procedures

Tester Note: This is a ‘shall’ statement – mandatory. It covers the full set of artifacts most relevant to quality assurance and testing.

- **Scenario ID: 3c**

  Notes & Best Practice Reference



Software peer reviews/inspections are recommended for all safety and mission-success related software. Detailed guidelines are contained in NASA-STD-8739.9.

- **Scenario ID: 4d**

  Benefits Table

Peer reviews/inspections provide the following additional benefits:

Useful for many types of products: documentation, requirements, designs, code	Simple to understand
Provide a way for sharing/learning of good product development techniques	Serve to bring together human judgment and analysis from diverse stakeholders in a constructive way
Can result in very efficient method of identifying defects early in the product's life cycle	Use a straight-forward, organized approach for evaluating a work product - To detect potential defects in a product To methodically evaluate each defect to identify solutions and track incorporation of these solutions into the product

Key benefits extracted:

- Useful for documentation, requirements, designs, code
- Simple to understand
- Knowledge sharing of good development techniques
- Brings together diverse stakeholder judgment constructively
- Efficient early defect detection in the life cycle
- Organized approach to detect, evaluate and track defect solutions

- **Scenario ID: 5e**

  Software Classification Matrix

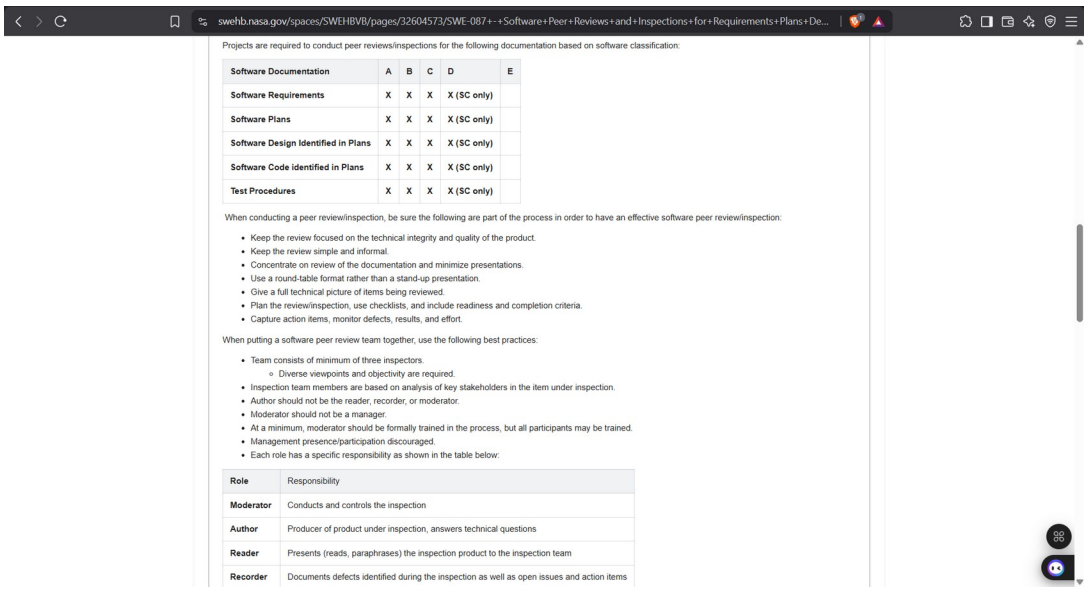
Projects are required to conduct peer reviews/inspections for the following documentation based on software classification:

Software Documentation	A	B	C	D	E
Software Requirements	X	X	X	X (SC only)	
Software Plans	X	X	X	X (SC only)	
Software Design Identified in Plans	X	X	X	X (SC only)	
Software Code identified in Plans	X	X	X	X (SC only)	
Test Procedures	X	X	X	X (SC only)	

X marks indicate mandatory peer review/inspection. For Class D the requirement is limited to Safety-Critical (SC) items only. Class E has no mandatory entries shown. Academic projects can treat themselves as Class C and still apply the full set of reviews.

- **Scenario ID: 6f**

  Roles Table



Defined roles:

- Moderator – Conducts and controls the inspection
- Author – Producer of the product; answers technical questions
- Reader – Presents (reads/paraphrases) the product to the team
- Recorder – Documents defects, open issues and action items
- Inspectors – Examine the product and identify defects (minimum 3)

Critical rule: Author must NOT be the Reader, Recorder or Moderator. Moderator should not be a manager.

- **Scenario ID: 7g**

  Inspection Process Steps

Software peer reviews are conducted using the following steps:

Step	Description
Planning	Organize inspection, inspection package contents, required support, schedule
Overview	Educational briefing at time of package distribution to explain materials at high level
Preparation	Inspectors individually look for and document defects and develop questions
Inspection Meeting	Inspectors examine the product as a group to classify and record defects, capture open issues and action items
Third Hour	Optional informal meeting to resolve open issues and discuss solutions
Rework	Author corrects major defects (others when cost and schedule allow)
Follow-up	Moderator verifies all major and other dispositioned defects have been corrected; no new defects introduced; all action items/open issues are closed

The formal process consists of:

1. Planning – Organize package, support, schedule
2. Overview – High-level educational briefing
3. Preparation – Individual defect search & questions
4. Inspection Meeting – Group classification & recording of defects
5. Third Hour – Optional informal resolution of open issues
6. Rework – Author corrects major defects
7. Follow-up – Moderator verifies corrections & closure

- **Scenario ID: 8h**

  Best Practices List

When conducting a peer review/inspection, be sure the following are part of the process in order to have an effective software peer review/inspection:

- Keep the review focused on the technical integrity and quality of the product.
- Keep the review simple and informal.
- Concentrate on review of the documentation and minimize presentations.
- Use a round-table format rather than a stand-up presentation.
- Give a full technical picture of items being reviewed.
- Plan the review/inspection, use checklists, and include readiness and completion criteria.
- Capture action items, monitor defects, results, and effort.

When putting a software peer review team together, use the following best practices:

- Team consists of minimum of three inspectors.
  - Diverse viewpoints and objectivity are required.
- Inspection team members are based on analysis of key stakeholders in the item under inspection.
- Author should not be the reader, recorder, or moderator.
- Moderator should not be a manager.
- At a minimum, moderator should be formally trained in the process, but all participants may be trained.
- Management presence/participation discouraged.
- Each role has a specific responsibility as shown in the table below:

Process best practices include: keep focus on technical quality, keep simple/informal, minimize presentations, use round-table format, plan with checklists and readiness/completion criteria, capture action items and metrics.

Team best practices: minimum three inspectors with diverse viewpoints, stakeholder-based selection, Author excluded from Moderator/Reader/Recorder roles, Moderator trained and not a manager, management participation discouraged.

- **Scenario ID: 9i**

Rationale for Early Reviews

> As documented in NASA-GB-8719.13, NASA Software Safety Guidebook [276] , "Formal Inspections have the most impact when applied early in the life of a project, especially the requirements specification and definition stages of a project. Impact means that the defects are found earlier, when it's cheaper to fix them....Formal Inspection greatly improves the communication within a project and enhances understanding of the system while scrubbing out many of the major errors and defects."

"Formal Inspections have the most impact when applied early in the life of a project, especially the requirements specification and definition stages of a project. Impact means that the defects are found earlier, when it's cheaper to fix them…. Formal Inspection greatly improves the communication within a project and enhances understanding of the system while scrubbing out many of the major errors and defects."

Tester Conclusion: This statement is the strongest justification for performing formal requirement reviews before design or coding begins. Defect cost multiplies dramatically the later it is found.

- **Scenario ID: 10j**

Walkthrough vs Formal Inspection – Comparison

Aspect	Walkthrough	Formal Inspection / Peer Review
Formality	Informal	Formal, process-driven
Leadership	Usually the author	Moderator (not the author)
Preparation	Light / optional	Mandatory individual preparation
Roles	Flexible	Defined (Moderator, Reader, Recorder, Inspectors)
Goal emphasis	Understanding + defects	Defect detection + process metrics
Checklist	Optional	Strongly recommended
Output	Notes / action items	Formal defect log + metrics

- **Scenario ID: 7g**

Summary of Findings (Software Tester View)

8. NASA SWE-087 makes peer reviews/inspections mandatory for requirements, plans, design, code and test procedures (class-dependent).
9. Five well-defined roles exist; Author is deliberately excluded from control roles to preserve objectivity.
10. A rigorous 7-step process ensures preparation, focused defect detection, rework and verified follow-up.
11. Early application (especially on requirements) yields the highest return by finding defects when they are cheapest to fix.
12. Best practices emphasize technical focus, checklists, metrics, diverse inspectors and exclusion of management from the meeting.
13. For Tutorial-3 the student should: (a) apply a requirements checklist to the project SRS, (b) simulate or perform a walkthrough and a formal inspection, (c) produce a defect log, and (d) reflect on the differences and value of early reviews.